A PRINCIPAL IDEAL DOMAIN IS NORMAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Cancelling a common factor in a fraction. Used at the start of the counted proof (lines 8289-8292). Reduces to the case of a coprime numerator and denominator.
- Integrality as a monic relation. Used in the counted proof at lines 8292-8294. Converts integrality of the fraction into a divisibility statement about the numerator.

Lemma 0.1. *A principal ideal domain is normal.*

Proof. Let R be a principal ideal domain. Let $g = a/b$ be an element of the fraction field of R integral over R. Because R is a principal ideal domain we may divide out a common factor of a and b and assume $(a, b) = R$. In this case, any equation $(a/b)^n + r_{n-1}(a/b)^{n-1} + \ldots + r_0 = 0$ with $r_i \in R$ would imply $a^n \in (b)$. This contradicts $(a, b) = R$ unless b is a unit in R. □

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